# Exact Nearest-Neighbor Search with Composable Lower Bounds: Self-Configuring Hierarchical Pruning, In RAM and Out of Core

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## ABSTRACT

Applications that cannot tolerate a missed true neighbor—deduplication, medical and legal retrieval, scientific analysis—need *exact* nearest-neighbor search, for which the standard tool is a cheap lower bound that prunes candidates without computing full distances. We develop such bounds from truncated PCA projections, which are tight precisely when variance concentrates in few dimensions, and observe that they are one member of a *family*: any region that provably contains a set of points yields an exact lower bound via the query-to-region distance, and different region shapes are tight for different data structures. We compose two complementary families—PCA truncation (tight for linearly correlated data) and triangle-inequality ball bounds around $k$-means centroids (tight for clustered data, and exact for *any* centers, so approximation quality affects only speed, never correctness)—into a single exact cascade. Because neither family helps on all data, the index *configures itself* from a sample: cascade widths are chosen from the eigenvalue curve (fixed widths tuned for 128-dimensional descriptors misread 1536-dimensional OpenAI embeddings as unprunable; widths placed at fixed variance targets recover them), and a three-tier diagnostic—eigen curve, beyond-covariance statistics, then a measured trial of the real cascade—decides which levels to build. We report results in hardware-independent work units alongside wall-clock, because wall-clock proved regime-dependent (the same algorithm measured 4.7×, 3.6×, 2.7×, and parity under changes of numeric type, machine load, and implementation tuning) while its work counts never moved. On four real datasets spanning 96–1536 dimensions (SIFT1M, GIST1M, DBpedia-OpenAI-1M, DEEP10M), exact 1-NN costs 3.9–20.8× less work than a scan, with wall-clock gains *growing* with dimensionality (2.6× at 128-D to 6.2× at 960-D): brute force pays the full dimension per point while the cascade pays for intrinsic structure. Exact $k$-NN generalizes the pruning radius to the $k$-th order statistic over evaluated points; it wins to $k \approx 10$ and honestly loses in RAM at $k = 100$. Out of core, the cascade's fetch set concentrates in few $k$-means clusters (27 of 1024 on the embeddings), and a measured two-tier layout under `O_DIRECT` cold reads answers exact queries 7–17× faster than a full disk scan while reading 40–730× fewer bytes. Against ADSampling, the state-of-the-art statistical distance-comparison pruner, our in-RAM work counts are mixed (we win at 96–128 dimensions, it wins on the embeddings), but its $\Delta d = 32$-dimension floor forces it to touch *every* vector, so it cannot leave RAM: deterministic bounds pay precisely where bytes are expensive. Exactness is verified on every query of every experiment. All code, diagnostics, and experiments are public.

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**PVLDB Artifact Availability:**
The source code, data, and/or other artifacts have been made available at https://github.com/taiwuchiang-gmail/hierarchical-pca-search.

## 1 INTRODUCTION

Finding the exact nearest neighbor in a large, high-dimensional dataset is bottlenecked by the cost of computing full distances. Approximate methods (LSH [7], HNSW [10], IVF [8]) trade recall for speed; for applications where a missed true neighbor is unacceptable—deduplication, medical and legal retrieval, scientific analysis—the alternative is exact pruning in the GEMINI tradition [4]: compute a cheap *lower bound* on each candidate's distance, and discard the candidate if the bound already exceeds the best distance found so far. Our starting point is such a bound from truncated PCA projections (§3.1): after rotating the data onto its principal axes, the distance between $k$-dimensional prefixes lower-bounds the true distance, so a cascade of progressively finer prefixes—seeded by a dynamic Top-$K$ probe—prunes over 99.9% of the SIFT1M dataset while returning provably exact results.[1]

That bound has a blind spot, and it is structural. PCA truncation is tight exactly when variance concentrates in few principal directions—linearly correlated data. Datasets whose structure is *clustered* rather than correlated can present a flat eigenvalue spectrum, collapsing the PCA cascade toward brute force, while remaining highly compressible in a different sense: points huddle near a modest number of centroids. No single bound family serves all data, and, worse, a practitioner has no way to know before building an index which family—if any—their data supports.

This paper makes two moves. First, **exact bounds compose**: the maximum of valid lower bounds is a valid lower bound, so


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[1] Preliminary versions of this work appeared as two self-archived reports (doi:10.5281/zenodo.21387358, doi:10.5281/zenodo.22199955). This paper merges, extends, and supersedes them: the $k$-NN generalization, the eigen-curve width selection, three of the four datasets, the measured out-of-core results, and the ADSampling baseline are new.

bound families with complementary blind spots can share one cascade. We add a triangle-inequality *ball bound* around $k$-means centroids—the clustered-data complement of PCA truncation—and show that it is exact for *any* choice of centers: clustering quality affects only pruning power, never correctness. Second, **the composition is chosen by measurement**: a three-tier diagnostic, run on a sample in seconds, rates the PCA levels from the eigenvalue decay, detects structure beyond the covariance with two cheap statistics, and—because we show statistics alone mispredict in both directions—settles the question with a measured trial of the real cascade, emitting an ADD/SKIP verdict per level.

There is also an economic motivation. An exhaustive flat scan requires the entire dataset resident in RAM—the most expensive byte in the machine, running roughly an order of magnitude above NVMe per byte, and more under supply pressure. The pruning cascade needs only its metadata resident: 8–40 bytes per vector versus 512 bytes per vector for `fp32` data. At one billion vectors this is 8–40 GB of RAM versus 512 GB—a commodity box versus a specialty one—with exactness preserved.

Our contributions:

1. **Exact hierarchical pruning from composable bounds**: a PCA-prefix cascade with a dynamic Top-$K$ pruning radius, unified with the triangle-inequality ball bound under a containing-region principle in which each bound family covers the other's structural blind spot (§3).
2. A **composed exact cascade**—ball level plus PCA levels—with a radius re-probe, generalized to exact $k$**-NN** via the $k$-th order statistic over evaluated points. Storage overhead is ~8 bytes per point (§5).
3. **Self-configuration that includes the cascade widths**: levels are placed at fixed cumulative-variance targets on the eigenvalue curve (a fixed ladder is silently calibrated to one dimensionality and misreads others), then the three-tier funnel—statistics nominate, measurement decides—rules on each level, with verdicts derived from measured work ratios; negative verdicts included (§4).
4. **Full-scale evidence on four real datasets** (96–1536 dimensions, $10^6$–$10^7$ points) in hardware-independent work units, with wall-clock as cost-model validation, negative controls, and sample-to-full-scale extrapolation checks (§6).
5. A **measured out-of-core design**: the fetch set's cluster locality, a two-tier on-disk layout, and cold-read benchmarks under `O_DIRECT` showing exact search 7–17× faster than a full disk scan (§6.6).
6. An **ADSampling baseline** in a shared work currency, locating the honest boundary between statistical and deterministic pruning (§6.7).

## 2 BACKGROUND AND RELATED WORK

**Lineage.** Neither bound family is new, and we do not claim otherwise. Lower-bounding by dimensionality reduction descends from the GEMINI framework [4] and the VA-file [13]. Pruning by the triangle inequality around reference points is the basis of ball trees [11], vantage-point trees [14], GNAT [1], and Elkan's accelerated $k$-means [3]. The approximate cousins are FAISS IVF [8] (the same $k$-means geometry without exactness) and OPQ [6] (a per-cell rotation, the ellipsoid rung of the shape ladder in §3). Disk-resident ANN systems such as DiskANN [12] and SPANN [2] target the same memory-economics regime we do, but are approximate. Closest in mechanism is ADSampling [5], which also evaluates incremental prefix distances after a rotation, but replaces the deterministic bound with a sequential hypothesis test under a *random* rotation (its layout-level successor is PDX [9]); we measure against it directly in §6.7. Our contribution is the *exact composition* of the two bound families in one cascade, the *measured self-configuration* (including the cascade widths themselves), and a diagnostic that also tells the practitioner when to walk away.

## 3 COMPOSABLE EXACT BOUNDS

### 3.1 The PCA Truncation Bound

Let $X, Y \in \mathbb{R}^d$ and let $X', Y'$ be their images under the orthogonal rotation onto the dataset's principal axes. Rotation preserves the squared Euclidean distance, $L_2^2(X,Y) = L_2^2(X',Y') = \sum_{i=1}^{d}(x_i' - y_i')^2$. Let $T_k$ retain the first $k$ coordinates.

Lemma 3.1. *For any truncation width $k < d$, $L_2^2\big(T_k(X'), T_k(Y')\big) \leq L_2^2(X,Y)$.*

Proof. The true squared distance is a sum of $d$ non-negative terms; the truncated distance is the sum of the first $k$ of the same terms, and $\sum_{i=k+1}^{d}(x_i' - y_i')^2 \geq 0$. □

If the prefix distance to a candidate already exceeds the current search radius $r$, the candidate is discarded without ever computing its full distance—the GEMINI no-false-dismissal condition [4]. Two choices make the bound an algorithm. First, *which rotation*: any orthogonal rotation satisfies Lemma 3.1; PCA makes the bound *tight*, since by the Karhunen–Loève theorem it packs the maximum variance—and hence, on average, the largest share of any pairwise distance—into the first $k$ coordinates (quantified in Eq. (3)). Exactness is unconditional; quality is not, a separation that recurs throughout this paper. Second, *where the radius comes from*: rather than a static threshold, a Top-$K$ probe at the coarsest width computes exact distances for the $K$ most promising candidates ($K = 50$) and sets $r$ to their minimum. Because the prefix distance is a lower bound, this radius is mathematically safe from the first comparison on, with no prior knowledge of the neighbor distance. A cascade of progressively finer prefixes then re-prunes the survivors at each width; on SIFT1M, widths $(8, 16, 32, 64)$ discard all but an average of 564 of $10^6$ vectors before any full-dimensional evaluation, and §4 replaces the fixed widths with ones derived from the data's own eigenvalue curve.

### 3.2 The Containing-Region Principle

Let $R \subseteq \mathbb{R}^d$ be any region that provably contains a set of points $C$. Then for every $p \in C$ and any query $q$,

$$\|q - p\| \;\geq\; \mathrm{dist}(q, R), \tag{1}$$

so $\mathrm{dist}(q, R)$ is an exact pruning bound for every member of $C$ at once. A bound family is therefore a *shape* family, and a shape is admissible when (a) containment is guaranteed and (b) $\mathrm{dist}(q, R)$ is

**Table 1: Shape families and their exact bounds. Each is tight for a different data structure; each mirrors a quantizer family.**

| Shape | Bound cost | Tight for | Quantizer twin |
|---|---|---|---|
| ball | $\lvert \lVert q-c\rVert - d_p \rvert$ | clusters | VQ |
| ellipsoid | per-axis scaled | stretched clusters | OPQ |
| slab | projection residual | local low rank | local PCA |
| cone×annulus | closed form | radial/gain data | gain–shape |

cheap. Table 1 lists the natural ladder. Each geometric family is the twin of a quantizer family: a quantizer's codeword-plus-distortion model *is* a containing region. We use the first two rungs; the others are available under the same principle.

### 3.3 The Ball Bound

Cluster the dataset with $k$-means. For each point $p$ assigned to centroid $c$, store the scalar $d_p = \lVert p - c\rVert$ once, offline. At query time, compute the $k$ centroid distances $\lVert q - c_j\rVert$ (with $k \ll N$), and the triangle inequality gives, for every point,

$$\lVert q - p\rVert \;\geq\; \big\lvert\, \lVert q - c_{a(p)}\rVert - d_p \,\big\rvert, \tag{2}$$

where $a(p)$ is $p$'s cluster. Evaluating (2) for all $N$ points requires one gather and one subtraction per point—*no distance computations.* A coarser, per-cluster form uses the cluster radius $R_j = \max_{p \in j} d_p$: if $\lVert q - c_j\rVert - R_j$ exceeds the current radius, the entire cluster is eliminated in one comparison, which in an out-of-core layout skips a whole contiguous block (§5.4).

### 3.4 Exactness Is Unconditional; Quality Is Not

Inequality (2) is the triangle inequality: it holds for *any* reference points whatsoever. $k$-means enters only because it minimizes the mean $d_p^2$, which is what makes the bound *tight.* Three corollaries follow. Stale centers remain correct as the dataset drifts (pruning degrades gracefully; re-cluster when a diagnostic says so). Insertion is $O(1)$: assign the new point to its nearest existing centroid and store $d_p$; no re-clustering is required for correctness. And the composed index inherits the PCA bound's property (§3.1) in mirror image: there, any orthogonal rotation is exact and PCA makes it tight; here, any centers are exact and $k$-means makes them tight. Since the maximum of lower bounds is a lower bound, the two families compose freely in one cascade.

### 3.5 Complementarity

PCA truncation is tight iff the eigenvalue spectrum decays steeply—a *global, linear* property (Figure 1, left). The ball bound is tight iff points sit close to centroids relative to the spread of $\lVert q - c\rVert$—a *local* property that holds for multimodal data and, more generally, for data of low intrinsic dimension, where $d_p$ and the centroid distances spread widely instead of concentrating. The blind spots are disjoint by construction. Our clearest demonstration: a synthetic dataset of 512 tight clusters whose centers span all dimensions has a nearly flat spectrum (95% of variance needs 111 of 128 dimensions; the PCA cascade manages 2.1×), yet the ball bound prunes to 3% before a single projection is computed.

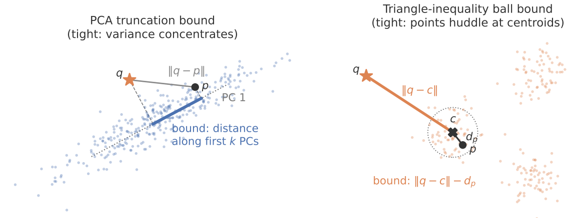


**Figure 1: The two exact bound geometries. Left: the PCA truncation bound is the distance between the projections onto the first $k$ principal components—tight when variance concentrates along few directions. Right: the ball bound $\lVert q - c\rVert - d_p$ from the triangle inequality—tight when points huddle near their centroids. Each is loose exactly where the other is tight.**

**Sample** — 100k of your vectors, seconds of compute, *before any index is built.*

**Tier 1 — eigenvalue curve.** Rates each PCA level's expected tightness and *picks the cascade widths*: one level at the dimension reaching each cumulative-variance target $\{30, 50, 70, 85\}$% (rounded to multiples of 8, capped at $d/2$).

*nominates*

**Tier 2 — beyond-covariance statistics.** Excess kurtosis of PCA components and sqdep against the $\sqrt{2/n}$ floor: does structure beyond the covariance *exist*? (Existence does not imply it is harvestable.)

*nominates*

**Tier 3 — measured trial.** Fit the *real* index on the sample; run the *real* cascade with the ball level on and off; measure the work ratio and the ball stage's own survivor rate.

*decides*

**PCA levels**
STRONG (> 8× work)
MODERATE (> 2×)
POOR → brute force / ANN

**Ball level**
ADD (measured gain)
SKIP (bound work cancels savings)

**Figure 2: The self-configuration funnel. Tiers 1–2 are statistics: cheap, and only allowed to *nominate* (tier 1 also derives the cascade widths from the eigenvalue curve). Tier 3 is a measured trial of the real cascade and alone *decides.* §6.5 shows the sample verdicts holding at $10^6$–$10^7$ scale; the tier-2 mispredictions below (SIFT and latent-8, in opposite directions) are why no statistic can replace the trial.**

## 4 THE SELF-CONFIGURATION FUNNEL

The funnel answers, from a sample and in seconds, "which levels should this index have for *my* data?" Figure 2 shows its shape: two statistical tiers that may only *nominate*, and one measured tier that alone *decides.*

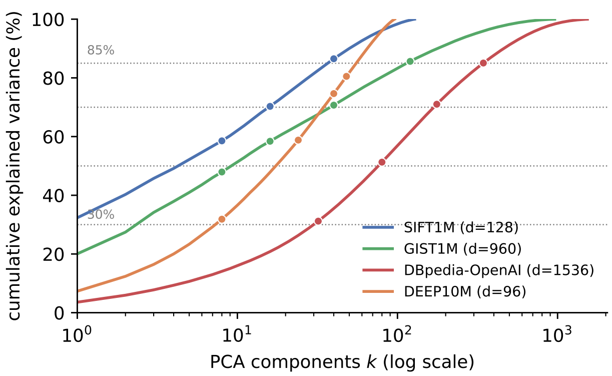


**Figure 3: Cumulative explained variance on 100k samples (log-scale components): each dataset's prunability signature, since by Eq. (3) the curve is the expected fractional tightness of a width-$k$ bound. Dots mark the widths the funnel selects at the $\{30, 50, 70, 85\}$% targets (dotted lines). SIFT and GIST are steep-headed; DBpedia's curve is shifted right—the reason a fixed $d$=128 ladder misreads it as unprunable; DEEP is flattest, matching its MODERATE verdict, and its last width sits below the 85% line because of the $d/2$ cap.**

**Tier 1 — eigenvalue decay, which also *picks* the widths.** Fit PCA on a sample. The eigenvalue curve is not merely a heuristic signal: in the PCA basis the coordinates of the difference between two independently drawn points are uncorrelated with variances $2\lambda_i$, so for typical candidate pairs

$$\frac{\mathbb{E}\, L_2^2\big(T_k(X'), T_k(Y')\big)}{\mathbb{E}\, L_2^2(X, Y)} = \frac{\sum_{i \le k} \lambda_i}{\sum_i \lambda_i}, \tag{3}$$

the cumulative explained variance at $k$. The curve therefore reads directly as the *expected fractional tightness* of the width-$k$ bound on the very population the cascade prunes—each dataset's curve is its prunability signature (Figure 3). Two blind spots follow from the same statement, and each motivates a later tier. Equation (3) is an expectation over pairs, while pruning is a tail event—a candidate is discarded only when its prefix distance clears the radius—so scalar summaries of the curve can mislead: GIST1M needs 303 of 960 dimensions for 95% variance, a "flat spectrum" by the usual standard, yet its steep *head* (48% by eight components) is what the early levels price, and the measured verdict is STRONG. And the spectrum is second-order only: clustered structure that no rotation can expose is invisible to it, which is what tier 2 screens for and tier 3 settles.

Preliminary versions of this work used a fixed ladder (8, 16, 32, 64)—which, we discovered, is silently calibrated to $d \approx 128$: on 1536-dimensional OpenAI embeddings it stops at 4% of the dimensions, prunes almost nothing (68% survivors after the 64-D stage), and the funnel misreads eminently prunable data as hopeless. The fix makes tier 1 constructive: place one level at the dimension reaching each cumulative-variance target $\{30, 50, 70, 85\}$%, rounded up to a multiple of 8 and capped at $d/2$. Measured against hand-tuned ladders on 100k samples (work ratio), the rule ties SIFT (8.2× vs 8.5×) and *wins* GIST1M (15.2× vs 9.3×) and DBpedia-OpenAI (8.2× vs 7.6×). A self-configuring index cannot have a hard-coded ladder; the eigen curve nominates the widths, and the trial below still decides.

**Verdicts are work ratios, not survivor fractions.** A second dimension-calibration bug, caught the same way: rating prunability by the final survivor fraction implicitly assumes one dimensionality. GIST1M's 6.9% finalists scored "poor" under a threshold tuned for SIFT—while meaning 9.3× fewer terms at $d = 960$, because a scan always pays $n \cdot d$. The verdict now derives from the measured work ratio (sample terms vs. $n \cdot d$), which is dimension-aware by construction.

**Tier 2 — beyond-covariance statistics.** Two cheap statistics on the PCA-rotated sample: the mean excess kurtosis of components (marginal non-Gaussianity), and sqdep, the mean $|\mathrm{corr}(z_i^2, z_j^2)|$ over component pairs (joint dependence that no rotation can remove; finite-sample noise floor $\approx \sqrt{2/n}$). These detect that structure beyond the covariance *exists*—not whether it is *harvestable*.

**Tier 3 — the measured trial.** Fit the real hybrid index on the sample, run the real cascade with the ball level on and off, and report the measured ratio and the ball stage's own pruning rate. The verdict is ADD or SKIP.

Tier 3 must be the decider, and the evidence is that the statistics mispredict in *both* directions. On SIFT, tier 2 fires (sqdep = 0.032, 7× the floor: real beyond-covariance structure) yet the trial says SKIP (1.0×): ball bounds prune only 50–59%, and the bound work cancels the savings. On a latent-dimension-8 Gaussian, tier 2 is clean (kurtosis $\approx$ 0, sqdep at floor) yet the trial says ADD (1.9×): low intrinsic dimension spreads $d_p$ widely, so the bounds bite (pruning to 23%) with zero multimodality. Across all datasets the measured value of the ball level is a monotone function of its own pruning rate—survivors of 3% → 4–14×; 15–23% → 2–3×; 50–59% → parity; 100% → 0.8× (a measured loss)—and the trial measures exactly this quantity (Figure 4). No statistic substitutes for it.

## 5 THE HYBRID ALGORITHM

### 5.1 Offline

Fit PCA (covariance accumulated in chunks, so no full `float64` copy of the dataset is materialized), select the cascade widths from the eigen curve (§4), and fit $k$-means with $k = \mathrm{clip}(\sqrt{N}, 64, 4096)$ (MiniBatch; 24 s on SIFT1M for $k$=1024). Store per point a cluster id (`int32`) and $d_p$ (`float32`)—8 bytes—plus the $k \times d$ centroids and per-cluster radii. A measured sweep over $k \in \{8, 64, 256, 1024, 4096\}$ on SIFT1M shows ball-stage survivors falling monotonically (81%→43%) while fit cost rises 4 s → 47 s; $\sqrt{N}$ sits at the knee of the gain-versus-fit-cost curve, which is why it is the default rather than a tuned constant.

### 5.2 Query Pipeline

(1) $k$ centroid distances ($k \cdot d$ operations).
(2) Ball bounds (2) for all $N$ points: one gather and one absolute difference each.
(3) **Probe**: exact distances to the PROBE= 50 points with the smallest ball bounds seed a safe initial radius $r$ (the Top-50 probe of §3.1, seeded by ball bounds instead of coarse projections).

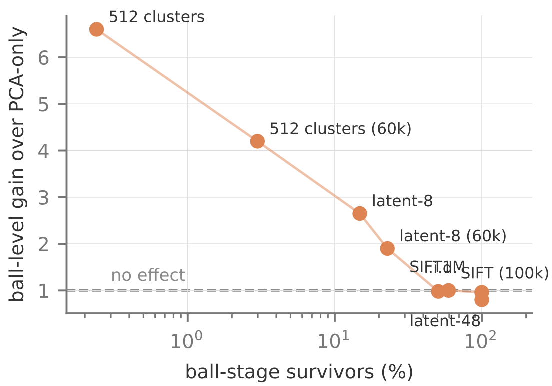


**Figure 4: The measured ball-value law across all eight datasets and trials (parity code): the ball level's wall-clock gain over the PCA-only cascade is a monotone function of its own pruning rate. The funnel's tier-3 trial measures precisely the $x$-axis quantity, which is why it predicts full-scale outcomes from a sample—including the negative verdicts below the dashed line.**

(4) Ball pruning: retain points with $\text{bound}^2 < r^2$.
(5) PCA cascade (8, 16, 32, 64) on the survivors, with a **re-probe** at the first PCA stage: before pruning, exact distances to the 50 best 8-dimensional candidates tighten $r$. Without this step the ball-seeded radius is loose and the cascade degrades (8-D survivors 27.2% versus 14.6% with it); with it, the cascade matches PCA-only tightness stage-for-stage.
(6) Full-distance check on the final survivors. The result is provably identical to brute force, and is additionally verified against the official ground truth in every experiment.

### 5.3 Exact $k$-NN

The pruning radius generalizes from the best evaluated distance to the $k$-th smallest exact distance among points evaluated so far: the $k$-th order statistic over a *subset* can only overestimate the true one, so any point whose lower bound reaches it provably cannot enter the $k$-NN set. One subtlety carries an exactness bug if missed: the evaluated multiset can contain a point twice (the probe and re-probe sets overlap), and the $k$-th order statistic of a multiset with duplicates is biased low—which *over*-prunes. The minimum (the 1-NN special case) is duplicate-invariant; the $k$-th order statistic is not, so the radius must be taken over *distinct* evaluated points. The probe size grows to $\max(50, k)$ so the initial radius is a valid $k$-th statistic. Correctness is verified against brute force across datasets, both cascade paths, and $k$ up to $n - 1$; a second subtlety surfaced in verification: integer-valued descriptors make exact distance *ties* real (SIFT query 13 has two points at squared distance exactly 61334, only one of which is in the official ground truth), so $k$-NN sets are unique only up to ties at the boundary.

### 5.4 Out-of-Core Layout

Pruning fragments I/O only if the layout ignores it—the classic index-versus-scan crossover: hundreds of scattered small reads can lose to a full sequential scan. Our two-tier design keeps *resident* only what pruning decisions touch: the rotated sketch of the first max(levels) PCA dimensions plus the ball metadata (`assign`+$d_p$, 8 bytes/point), centroids, and per-cluster offsets—2–8× smaller than the data. Full-dimensional vectors live on disk in two interchangeable files: original order (each fetch is one small aligned read) and *cluster-major* order, in which each $k$-means cluster is a contiguous, page-aligned block (<0.4% padding), written sequentially into `fallocate`-preallocated extents. The entire cascade—probe, re-probe, all pruning stages—runs on the resident tier; the disk is touched only by the fetch set (probe + re-probe + finalists), whose exact membership §6.6 measures. Two structural facts make the layout matter. First, the cluster-level bound alone is a weak block-skipper (on SIFT1M, 40–52% of clusters survive it). Second, the fetch set is far more concentrated than uniform—nearest neighbors of a query are near each other, so they share clusters—which converts the final fetch into few large reads. This is the wisdom of the VA-file [13], SPANN [2], and DiskANN [12]: sequential sketches, layout-aware fetches. The same $k$-means that supplies the bound supplies the layout—and §6.6 shows it earns its keep as a *partitioner* even on datasets where the funnel rejects it as a *pruning level*.

## 6 EVALUATION

**Metrics policy.** Primary claims are stated in hardware-independent work units, because wall-clock proved regime-dependent: the same algorithm measured 4.7×, 3.6×, 2.7×, and parity under changes of numeric type, machine load, and implementation tuning, while its work counts never moved (details below). The primary metrics are (1) full-vector fetches per query (the out-of-core I/O currency), (2) total bytes touched per query, *charging the pruning machinery itself*, (3) resident RAM bytes per vector, and (4) the survivor cascade, which is the algorithmic fingerprint. Wall-clock appears as validation of a first-order cost model ($t \approx$ bytes/bandwidth + ops/throughput), with FAISS supplying the tuned-constants reference—so a reader can plug their machine's constants into our work numbers and predict their own outcome.

**Setup.** Intel i9-9900K (8C/16T, AVX2+FMA, no AVX-512), dual-channel DDR4, 30 GB usable RAM; NVMe SSD measured under `O_DIRECT` at 2.04 GiB/s sequential (8 MiB reads) and 80 µs per random 4 KiB read; `faiss-cpu` 1.15 dispatching its AVX2 build; NumPy and OpenBLAS at the AVX2 tier—baseline and FAISS use the same instruction set. SIFT1M in-RAM timings (Table 2) are medians of 5 repetitions on a quiet machine with warm cache; the multi-dataset, $k$-NN, and out-of-core tables are medians or means over 100 queries in a single repetition on a quiet machine (work counts and byte counts are exact and run-invariant; only the millisecond columns carry run noise). Exactness is verified on every query of every experiment—against official ground truth where one exists, against in-harness brute force otherwise. Results are CPU-governor-insensitive (medians within 2%).

**Table 2: SIFT1M, 100 official queries, medians of 5 repetitions, implementation parity. All methods return the ground-truth neighbor on every query.**

| Method | ms/query | Speedup | Survivors per stage (%) |
|---|---|---|---|
| brute (NumPy) | 167.3 | 1.0× | – |
| PCA-only | 61.1 | 2.7× | 14.7 / 5.2 / 0.8 / 0.06 |
| hybrid | 62.2 | 2.7× | ball 50.6 → 14.6 / 5.1 / 0.7 / 0.05 |
| FAISS-flat, 1 thread | 29.3 | 5.7× | (scans 100%) |
| FAISS-flat, batch-100 | 5.9 | 28× | (scans 100%) |

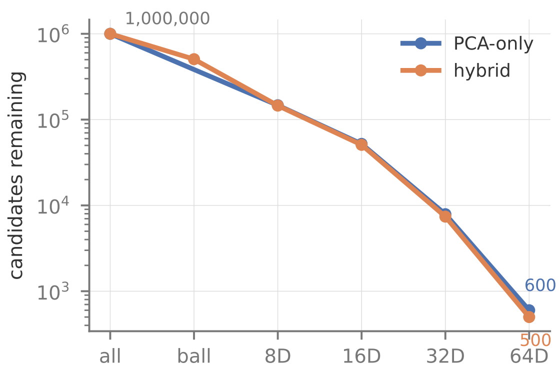


**Figure 5: Candidates remaining after each stage on SIFT1M (log scale, average over 100 official queries). The cascade discards four orders of magnitude before any full-distance evaluation; the hybrid enters the PCA stages with half the population and finishes equally tight. This profile is the algorithmic fingerprint: it was invariant under every change of numeric type, machine load, and implementation tuning in this paper.**

**Table 3: Hardware-independent work per query on SIFT1M. Terms are per-dimension $(x-y)^2$ evaluations; bound ops are per-point ball-bound evaluations, charged separately. Counts derive analytically from the survivor statistics (`count_ops` in the repository)—an exhaustive scan is $N \cdot d$ terms by definition, so no instrumentation of compiled libraries is needed. These columns are identical on every machine and every run.**

| Method | Terms (M) | Bound ops (M) | Terms vs brute |
|---|---|---|---|
| brute / FAISS-flat | 128.0 | – | 1.0× |
| PCA-only | 12.60 | 0 | **10.2×** |
| hybrid | 8.67 | 1.00 | 14.8× |

**Table 4: The exactness–speed–work–memory trade-off on SIFT1M (fp32, $10^6 \times 128$). Flat scans must hold every vector resident (512 B/vector); the cascades keep only pruning metadata resident and page full vectors from storage (§5.4): the 8-dimensional sketch costs 32 B/vector, the ball metadata 8 B/vector. Parentheses give the RAM saving relative to a flat scan. The last row situates disk-resident ANN systems: their residency (taken from their own reports, not measured here) is comparable to the cascades', but they answer the approximate problem—the exact-at-cascade-residency quadrant is the one this work occupies.**

| Method | Exact | ms/query | Terms (M) | Resident RAM |
|---|---|---|---|---|
| FAISS-flat, batch-100 | yes | 5.9 | 128.0 | 512 MB (1×) |
| FAISS-flat, 1 thread | yes | 29.3 | 128.0 | 512 MB (1×) |
| brute (NumPy) | yes | 167.3 | 128.0 | 512 MB (1×) |
| PCA-only | yes | 61.1 | 12.6 | 32 MB (16×) |
| hybrid | yes | 62.2 | 8.7 +1.0 | 8–40 MB (13–64×) |
| IVF / disk ANN [2, 8, 12] | no | – | – | ~8–64 MB |

## 6.1 SIFT1M

Table 2 gives the in-RAM picture, and Figure 5 the cascade behind it. The PCA cascade alone reaches 2.7×; **the ball level is a wall-clock tie on SIFT**—its 50% pruning saves about what its bound computation costs—exactly as the sample trial predicts (SKIP, §4). The hybrid's SIFT-side value is therefore out-of-core and residency only (§5.4); the wall-clock case for the ball level rests on clustered and low-intrinsic-dimension data (§6.4). The survivor cascade exactly reproduces the preliminary reports'.

Table 3 states the same experiment in the primary currency. Three things the wall-clock hid become explicit. First, the cascade's *algorithmic* reduction is 10.2×—machine-free, and invariant across every run in this paper, where the millisecond columns wobbled with load. Second, FAISS-flat performs 10–15× more work than the cascade and still wins in-RAM wall-clock: the engineering-versus-algorithm separation is now quantitative rather than rhetorical. Third, the hybrid cuts terms further (12.6 → 8.7 M) but adds 1.0 M bound operations of a different mix (gathers rather than streaming reads)—which is why its wall-clock ties PCA-only despite the smaller term count, and why work accounting that omits the pruning machinery would mislead.

FAISS `IndexFlatL2` wins the in-RAM race outright, scanning all $10^6$ vectors in 29.3 ms single-threaded. The gap is fusion, not SIMD: both sides run AVX2, but FAISS's kernel makes one pass (~512 MB of traffic per query) where NumPy's expression materializes temporaries (~3 passes, 1.5–2 GB), a 5.6× advantage at identical work. The comparison scopes the claim rather than defeating it: the cascade touches 1,700× fewer full vectors, requires 8–40 resident bytes per vector against 512 for any flat scan, and its bounds compose with FAISS-grade kernels rather than competing with them (the cascade could drive such a kernel over its survivor sets). Disk-resident ANN systems occupy the same economics but surrender exactness. Table 4 summarizes the resulting three-way trade among speed, work, and the machine's most expensive resource.

**Why wall-clock is the wrong headline (measured).** Two confounds were caught only by interrogating ratio changes. *Numeric type*: sklearn's PCA silently outputs `float64`, whose bandwidth-bound brute baseline is 1.86× costlier; a 2 × 2 decomposition on

**Table 5: Exact 1-NN at full scale, eigen-curve-selected levels, PCA-only cascade. Terms count per-dimension $(x-y)^2$ evaluations; a scan is $n \cdot d$ by definition. All queries exact.**

| Dataset | $n$ | $d$ | Wall | Terms (M) | Work red. |
|---|---|---|---|---|---|
| SIFT1M | $10^6$ | 128 | 2.6× | 12.6 | 10.2× |
| GIST1M | $10^6$ | 960 | **6.2×** | 46.1 | **20.8×** |
| DBpedia-OpenAI | $10^6$ | 1536 | 6.0× | 119.6 | 12.8× |
| DEEP10M | $10^7$ | 96 | 1.2× | 244.0 | 3.9× |

**Table 6: Exact $k$-NN, PCA-only cascade, wall-clock speedup over brute force and (in parentheses) work reduction in terms. All queries exact; $k$=100 additionally matches official ground truth up to boundary ties.**

| Dataset | $k = 1$ | $k = 10$ | $k = 100$ |
|---|---|---|---|
| SIFT1M | 2.6× (10.2×) | 1.9× (6.2×) | 0.7× (1.6×) |
| GIST1M | 6.2× (20.8×) | 4.0× (10.3×) | 0.9× (1.5×) |
| DBpedia-OpenAI | 6.0× (12.8×) | 3.1× (5.8×) | 0.8× (1.2×) |
| DEEP10M | 1.2× (3.9×) | 0.8× (2.0×) | 0.3× (0.6×) |

identical code measures the same cascade at 3.6× (f64—the first preliminary report's regime, matching its reported 4.3×) versus 2.7× (f32). *Implementation parity*: an unnecessary full-array gather in our own PCA-only path inflated the ball level's apparent contribution to 2.0×; fixing it revealed the tie in Table 2. Machine load adds a further multiplier: background bandwidth contention slows the scan-bound baseline most, flattering all pruners. Survivor counts were invariant through every one of these changes. Hence the metrics policy.

## 6.2 Four Real Datasets: the Gain Grows with Dimension

Table 5 answers the question SIFT alone cannot: does anything beyond SIFT prune? Everything measured does, and the wall-clock gain *grows* with dimensionality—2.6× at 128-D, 6.2× at 960-D, 6.0× at 1536-D—because brute force pays the full dimension for every point while the cascade pays for intrinsic structure. The result that mattered most in prospect: modern learned embeddings (DBpedia abstracts under OpenAI `ada-002`), whose flat spectra (664 of 1536 dimensions for 95% variance) looked like the approach's natural failure mode, prune to 0.64% finalists once the cascade widths respect the eigen curve (§4)—with the *fixed* (8, 16, 32, 64) ladder the same data measures 1.3× and reads as unprunable. DEEP10M is the honest boundary case: 96 dimensions of already-PCA-compressed deep features leave a 3.9× work reduction but only 1.2× in-RAM wall-clock; the funnel's sample verdict (MODERATE, "useful out-of-core") says exactly this in advance, and §6.6 confirms where its value lives. On all four datasets the ball level's sample verdict (SKIP everywhere; survivor rates 50–99%) was confirmed at full scale—the ball-value law of §4 holding on real data at $10^6$–$10^7$ points.

**Table 7: Structure regimes (synthetic, $n$ = 200k, $d$ = 64 for the first two; $n$ = 60k, $d$ = 128 for the funnel rows), implementation parity. "Ball survivors" is the ball stage's own pruning rate—the quantity the funnel's trial measures.**

| Dataset | PCA-only | Hybrid | Ball surv. | Verdict |
|---|---|---|---|---|
| 512 tight clusters | 2.1× | **13.9×** | 3% | ADD |
| latent-8 Gaussian | 2.1× | 5.6× | 15–23% | ADD |
| SIFT1M | 2.7× | 2.7× | 50–59% | SKIP |
| latent-48 Gaussian | 1.9× | 0.8× of PCA | 100% | SKIP |
| i.i.d. Gaussian | 0.3× | — | 100% | SKIP (all) |

## 6.3 Exact $k$-NN

Table 6 reports the $k$-NN generalization (§5.3) with deliberate honesty: the method wins to $k \approx 10$ and *loses* in-RAM wall-clock at $k = 100$ on every dataset. The mechanism is structural, not incidental: the pruning radius is the $k$-th best distance, which grows with $k$, and survivor counts grow with it—at $k$=100 the ball stage passes 99–100% and the cascade approaches a scan with overhead. The $k$=100 work reductions (1.2–1.6× on the million-scale sets) still translate to fewer bytes fetched out of core, which is where large-$k$ exact queries should run (§6.6).

## 6.4 Structure Regimes

Table 7 spans the design space, negative results included deliberately. On the flat-spectrum clustered dataset—PCA's blind spot—the ball level prunes to 3% before any projection is computed and the hybrid reaches 13.9×; its bound requires no multiplications, so it undercuts even the 8-dimensional scan. On the latent-48 blob the ball level is measured to *hurt* (0.8×: 100% survivors, pure overhead), and on i.i.d. noise the diagnostic correctly reports that nothing works and brute force or ANN should be used. The i.i.d. row also shows the PCA cascade itself losing to brute force (0.3×)—the honest boundary of the entire approach.

## 6.5 Extrapolation

The funnel's trial on a 100k SIFT sample returns 1.0× (SKIP); the full-1M parity benchmark measures exactly that (61.1 versus 62.2 ms). The check extends to every dataset: GIST1M sampled at 9.3× work with the fixed ladder and 15.2× with eigen-curve levels; the full-1M runs measured 13.1× and 20.8× respectively (the sample is conservative). DEEP10M sampled MODERATE at 3.6× and measured 3.9× at $10^7$ points; the ball level's sample SKIP was confirmed at full scale on all four datasets (e.g. GIST: 92% ball survivors, identical term counts, strictly worse wall-clock). The diagnostic predicts full-scale outcomes from a sample—including the negative verdicts, which is its most valuable behavior: it prevents building a useless level.

## 6.6 Out of Core: Measured Cold Reads

Section 5.4 motivated the two-tier layout from structure; Table 8 measures it end to end. The protocol: all reads use `O_DIRECT` with page-aligned buffers, so the page cache is bypassed and every query pays device prices—a cold-cache protocol with no cache-flushing

**Table 8: Cold-read exact 1-NN under `O_DIRECT` (every byte from the device; page cache bypassed, so each query is cold by construction). Scan streams the whole file; point fetches each needed vector from the original-order file; block fetches whole clusters from the cluster-major file. All arms return identical exact results.**

| Dataset | Arm | ms/query | MB/query | Reads |
|---|---|---|---|---|
| SIFT1M | scan | 402 | 488 | 62 |
| (488 MB) | point | **61** (7×) | **2.3** (212×) | 588 |
| | block | 81 (5×) | 37 | 73 |
| GIST1M | scan | 3006 | 3663 | 458 |
| (3.6 GiB) | point | **421** (7×) | **91** (40×) | 12,379 |
| | block | 751 (4×) | 1224 | 158 |
| DBpedia | scan | 5385 | 5860 | 733 |
| (5.7 GiB) | point | **318** (**17**×) | **8.0** (**730**×) | 1028 |
| | block | 418 (13×) | 374 | 36 |
| DEEP10M | scan | 3574 | 3664 | 458 |
| (3.6 GiB) | point | 1542 (2×) | **209** (18×) | 50,356 |
| | block | **1378** (3×) | 780 | 223 |

ceremony, deliberately conservative on NVMe, where cheap random reads and high sequential bandwidth both favor the *scan* baseline.

Three findings. **First, the headline**: on the OpenAI embeddings, exact cold-read 1-NN takes 318 ms against a 5.4-second scan (17×), reading 8 MB instead of 5.9 GB (730× fewer bytes). The byte ratios are device-independent; on storage slower or more seek-bound than NVMe they translate to larger time ratios. **Second, the fetch set is highly cluster-local**: the embeddings' ~1,014 fetched points occupy a median of 27 of 1024 clusters (uniform placement would touch ~644); SIFT's 583 points occupy 52. Nearest neighbors of a query are near each other. This is what makes the cluster-major arm viable at all—and it means the $k$-means structure earns residence as a *partitioner* even where the funnel rejects its *bound* (SKIP on all four datasets): the same geometry, useless for compute, valuable for layout. **Third, point-versus-block is a device property**: with serial reads at 80 µs the point arm's few thousand small fetches beat the block arm's fewer-but-heavier reads on three of four datasets, and only DEEP10M's 50k-point fetch set flips the order. Parallel small reads (NVMe queue depths reward exactly this) would extend the point arm's advantage; seek-bound media would flip every row toward block. The layout writes both files; the diagnostic should pick per device.

One design lesson surfaced only here: *the optimal cascade depth is deeper out of core than in RAM.* SIFT's eigen-curve levels (8, 16, 40) minimize in-RAM terms but fetch 3,381 points; the deeper (8, 16, 32, 64) ladder spends more sketch terms to fetch only 588—and wins the cold-read benchmark (61 vs 91 ms). In-RAM, terms are the currency; out of core, fetched bytes are, and the level-selection objective should follow the medium.

## 6.7 Baseline: ADSampling

ADSampling [5] is the natural adversary: it also prunes with incremental prefix distances after a rotation, but its rotation is random

**Table 9: ADSampling (faithful NumPy port of the reference implementation, $\epsilon_0$=2.1, $\Delta d$=32) versus our PCA-only cascade, $k$ = 1. Work in millions of dimension-terms per query; recall under float64-verified ground truth. Ours is exact by construction.**

| Dataset | ADS terms (recall) | Ours (exact) | Winner |
|---|---|---|---|
| SIFT1M | 33.0 (1.000) | **12.6** | ours 2.6× |
| GIST1M | **44.5** (1.000) | 46.1 | tie |
| DBpedia-OpenAI | **49.8** (1.000) | 119.6 | ADS 2.4× |
| DEEP10M | 325.8 (1.000) | **244.0** | ours 1.3× |

(partial distances become unbiased samples) and its stopping rule is a sequential hypothesis test with tunable error rate $\epsilon_0$—harder pruning per dimension, bought with the exactness guarantee. We ported the reference implementation to the same harness and currency (dimensions touched = terms) for parity. Table 9 is honestly mixed in-RAM: we win at 96–128 dimensions, ADS wins on the 1536-D embeddings, GIST ties; at $k$=10 ADS widens its advantage. On recall we correct our own first measurement: an earlier float32 ground-truth check miscounted near-duplicate points (catastrophic cancellation) as recall failures; under float64 verification ADS at default $\epsilon_0$ achieved perfect recall on every $k$=1 query we ran, with one true neighbor missed in 600 query-tasks (GIST, $k$=10, recall 0.9990). The distinction that matters is therefore *guarantee versus empirical*, not observed failure—and it is structural, since no $\epsilon_0$ makes the test a bound.

The decisive difference is not in RAM. ADS pays its $\Delta d$ = 32 floor for *every* point in the dataset—SIFT's 33.0M and DEEP10M's 325.8M terms sit exactly at 32$n$—so it must touch every vector, and out of core it reads full-scan bytes regardless of parameters (its successor PDX [9] restructures the layout dimension-major but still visits every vector). Our cascade fetches 0.06–1% of vectors (Table 8). The methods also compose rather than compete: a statistical test could accelerate the final full-distance checks over our survivor sets, at the usual price.

# 7 LIMITATIONS

High *intrinsic* dimensionality defeats every cheap bound: distance concentration flattens the eigenvalue spectrum and the spread of $d_p$ simultaneously, and the funnel exists precisely to detect this and say so (Table 7, last row); DEEP10M sits near this boundary in RAM. In-RAM wall-clock at native-code constants belongs to fused exhaustive scans (FAISS) at the 128-D scale, and ADSampling's statistical pruning beats our deterministic bounds in-RAM on the highest-dimensional embeddings (§6.7). Exact $k$-NN loses in-RAM wall-clock by $k$ = 100 on every dataset we measured (§6.3). The out-of-core measurements use `O_DIRECT` on datasets that fit in RAM—valid device-price accounting, but a larger-than-RAM run (with parallel rather than serial fetch I/O) remains future work, as do updates: the cluster-major layout assumes a static, rebuild-periodically index, the same assumption disk-ANN builds make. Ball-bound and layout quality depend on the $k$-means sample as any learned structure does, though correctness never does.

## 8 CONCLUSION

Exact lower bounds are shapes; shapes compose; and the right composition is a property of the data, measurable on a sample before any index is built. On that principle this paper delivers an exact $k$-NN index that configures itself end to end—cascade widths from the eigenvalue curve (without which 1536-dimensional embeddings masquerade as unprunable), levels from a measured trial whose sample verdicts are confirmed at $10^6$–$10^7$ scale on four real datasets, honest breaking points stated ($k = 100$ in RAM; i.i.d. data everywhere)—and an out-of-core argument that has become a measurement: exact cold-read queries 7–17× faster than a disk scan at 40–730× fewer bytes, against a state-of-the-art statistical pruner that structurally cannot leave RAM. The recurring lesson of this work is that every constant we had silently tuned to one dataset—cascade widths, verdict thresholds, even our own ground-truth checker—failed on the next one, and every fix was the same: derive it from a measurement. Measure the data to choose the index; count the work to predict the time.

**Artifact.** Code, diagnostics, and all experiment scripts: `github.com/taiwuchiang-gmail/hierarchical-pca-search`

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